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The Small Choroidal Melanoma: Diagnosis, Treatment Outcomes, and Unending Controversy

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Abstract

Small choroidal melanoma remains a diagnostic challenge due to its clinical overlap with choroidal nevus. This overlap encompasses a wide clinical phenotype of lesions termed indeterminate choroidal lesions. Treating an indeterminate lesion is weighed against significant visual morbidity from local treatment versus increased metastatic risk with tumor size. This comprehensive review includes diagnosis, controversies, management, and future therapy options for small choroidal melanoma.

Keywords

small choroidal melanoma; indeterminate choroidal lesion; uveal melanoma; ocular oncology; ocular tumor

Introduction

Uveal melanoma (UM) is the most common primary intraocular malignancy with the large majority presenting in the choroid.[1] Approximately 50% of patients will develop metastatic disease despite excellent local tumor control, and historical median overall survival of metastatic disease is approximately 1 year.[2] Local therapy options with possibly significant visual morbidity include radioactive plaque brachytherapy (PBT), proton beam or charged particle radiation, transpupillary thermotherapy (TTT), photodynamic therapy, local excision, and enucleation depending on clinical characteristics. Very small, flat nevi without any high-risk features are easy to diagnose clinically, while the Collaborative Ocular Melanoma Study (COMS) reported over a 99% diagnosis accuracy based on clinical exam for the medium- and large-sized tumors.[3] Small choroidal melanomas (SCM), however, remain a diagnostic challenge due to a clinical overlap with choroidal nevi. The overlap between a choroidal nevus with high-risk features and a definitive malignant choroidal melanoma encompasses a wide clinical phenotype of lesions that we term as indeterminate choroidal lesions (ICL) (Figure 1). Further, terming a lesion a melanoma often requires consideration for definitive treatment which is weighed against both significant visual morbidity from local therapy options and concern for increasing

metastatic risk with tumor size. Because of these uncertainties, small choroidal melanoma remains a controversial topic but with promising diagnostic and treatment modalities on the horizon. The primary purpose of this review is to discuss the significant progress made in understanding the molecular pathophysiology, imaging and biopsy findings, and less invasive treatment options for small choroidal melanoma.

Name Changes and Classification Systems

Difficulties to properly classify small choroidal melanomas lie with its significant clinical overlap with choroidal nevi as well as the local visual morbidity imposed with treatment. Donald Gass, in his 33rd Edward Jackson Memorial Lecture, rightly acknowledges the diagnostic challenges with the topic and offered several clinical characteristics that are still relevant today.[4] Diagnostic uncertainty has likely contributed to the various names applied to these lesions.[5] Such terms include indeterminate melanocytic proliferation, indeterminate choroidal lesion, atypical nevus, suspicious nevus, benign melanoma, simple melanoma, dormant melanoma, nevoma, and large nevus have been used, but these terms may be misleading.[6], [7] Singh and Grossniklaus astutely point out that a clinician may be biased in calling these lesions a choroidal nevus rather than small choroidal melanoma when management is observation so as not to appear they are delaying treatment for malignancy. [5] The authors suggest the lesions that are uncertain in diagnosis or share features that can be attributed to both a benign choroidal nevus or a small choroidal melanoma be called indeterminate melanocytic tumors (IMT) in order to add transparency to the clinical management.[5]

Various classification systems are used for choroidal melanoma which highlight the lack of uniformity in defining smaller lesions. The COMS Small Tumor Study, an observation-only study, included lesions that were between 1 mm and 3 mm in apical height and between 5 mm and 16 mm in largest basal diameter (LBD).[8] The 4th edition of the World Health Organization (WHO) classifies a choroidal nevus as usually less than 5 mm in diameter and less than 2 mm in tumor thickness.[9] Finally the Cancer Staging Manual of the American Joint Committee on Cancer (AJCC 8th edition), which assigns tumor size criteria by both tumor thickness and maximal basal diameter, does not provide a distinction between nevus and choroidal melanoma but instead defines T1 melanoma as tumor base less than or equal to 9 mm with thickness less than or equal to 6 mm, or tumor base 9.1–12 mm with thickness less than or equal to 3 mm.[10]

Epidemiology

As stated previously, uveal melanoma is the most common primary intraocular malignancy. [1] It is however much rarer than cutaneous melanoma and represents about 3% - 5% of all melanomas with a stable incidence of approximately five per million in the United States.[11], [12] Incidence varies by race with non-Hispanic Whites (6.02 per million) having a higher incidence than Hispanics (6.02 per million), African Americans (0.31 per million), and Asians (0.39 per million).[1], [13] These variations by race are consistent across the globe with lower rates in African and Asian countries compared to Caucasian countries.[14] Prior reports suggest there is an increased incidence in Caucasian patients from higher latitude countries.[14], [15] In the United States, males (5.8 per million) have a

higher incidence than females (4.4 per million) by about 30%.[11], [16] The median age at diagnosis is between 60 and 65 years.[17], [18] Increasing age is a risk factor for UM that peaks at about 70 years but then plateaus after 75 years.[14]

Choroidal nevus is the most common benign intraocular tumor, with a prevalence from the National Health and Nutrition Examination Survey (NHANES) database with retinal imaging of about 5% in adults over the age of 40.[19], [20] This imaging only included two 45-degree photographs centered on the fovea and optic disc, and the authors suggest the prevalence could be higher as much as 20–25%.[20] Contrary to ocular melanocytosis or melanocytomas, choroidal nevi are not considered congenital and are typically not seen at birth.[4] Nevi are uncommonly seen under the age of 18, and the prevalence increases with age.[21] In an epidemiological cross-sectional study of 5,533 children, 48 children exhibited a choroidal nevus and none underwent malignant transformation.[21] Using the Surveillance, Epidemiology, and End Result (SEER) database, Singh et al. estimated that the annual rate of malignant transformation of a nevus is 1 in 8845 in the U.S. White population suggesting that the vast majority of nevi do not transform into a melanoma.[22] Because of the significant clinical overlap between choroidal nevi with concerning features and small choroidal melanoma, the true incidence of small choroidal melanoma is difficult to quantify. Most tumors in the COMS small choroidal melanoma arm displayed clinical characteristics more consistent with choroidal nevi.[8], [23] The majority of the tumors did not grow during the first 5 years (69%). Most lesions were greater than 2 mm from the optic nerve (83.5%), did not have subretinal fluid (87.8%), or orange pigment (77.7%).[8] Kujala et al. analyzed melanomas from 5 ocular oncology centers and categorized 24% as T1 (less than 3 mm in thickness and 12 mm in largest basal diameter).[10]

Diagnosis

Clinical Diagnosis: Symptoms for a choroidal melanoma can vary depending on tumor location and include blurred vision (38%), photopsia (9%), floaters (7%), visual field loss (6%), visible tumor (3%), pain (2%), metamorphopsia (2%), and other (3%).[24] Notably, 30% of patients are asymptomatic and the tumor is incidentally found on examination.[24] Most choroidal nevi do not cause symptoms unless there is associated macula-involving subretinal fluid. Typically, it is the growth rate of an ICL, compared to a benign choroidal nevus, that may suggest malignant transformation and thus most compatible with a clinical diagnosis of a uveal melanoma.[6] Nevi, while generally considered to be stable lesions, can have a slow growth rate in largest basal diameter. Mashayekhi et al. observed 284 nevi without malignant transformation and noted a median growth rate of 0.06 mm/year (mean, 0.06 mm/year, range, 0.01 – 0.36 mm/year).[25] Multiple studies have demonstrated a much faster growth rate for small UM by imaging, ranging from 1 mm/year to 1.8 mm/year.[6], [26] Prior studies using indirect ophthalmoscopy noted a more variable mean rate of 0.4 mm/year[27] and 3.4 mm/year.[28] Risk factors known for growth and therefore malignant transformation are similar to those described by Dr. Gass more than 40 years ago and include thickness greater than 2 mm, orange lipofuscin pigment, low ultrasound reflectivity, subretinal fluid, proximity to fovea and/or optic disc, presence of symptoms, and angiographic activity such as “hot spots” and irregular tumor vessels.[4], [6] These risk factors predictive of growth have been corroborated by multiple groups over the years.[26],

[29], [30], [31], [32] Dalvin et al. highlighted six risk factors predictive of growth into melanoma including tumor thickness > 2 mm, subretinal fluid, visual acuity loss of 20/50 or worse, orange pigment, hollow acoustic density, and tumor LBD > 5 mm.[33] 5-year estimated growth rate was 1% with no risk factors, 11% with one, 22% with two, 34% with three, 51% with four, and 55% with five factors.[33]

Multimodal imaging is therefore extremely valuable for ICL characterization and provides the clinician a composite of useful information when monitoring these lesions (Figures 2 and 3). Fundus photography allows for serial documentation and accurate monitoring of tumor margins to detect growth. Utilizing blood vessels as landmarks may aid the clinician when comparing sequential scans. Fundus autofluorescence helps assess retinal pigment epithelium integrity. Confluent hypo-autofluorescence on the tumor apex may correlate to areas of RPE atrophy, a sign of chronicity and possibly a very slow interval of growth. Focal areas of hyper-autofluorescence overlying the lesion and orange pigment on fundus photography or exam may correlate with lipofuscin accumulation.[19], [34] Confluent areas of hyper-autofluorescence on and around the tumor may also indicate the presence of current or prior subretinal fluid.

Enhanced Depth Imaging - Optical coherence tomography (EDI-OCT) helps confirm the tumors location within the choroid and its relation to the rest of the choroid, the RPE, and the overlying retina. The morphology of melanocytic lesions on OCT are typically smooth and dome-shaped, in contrast to metastatic lesions which may demonstrate RPE undulations and highly reflective points within the serous detachment.[35] OCT can help demonstrate the effect of the lesion on the overlying neurosensory retina and RPE including intraretinal fluid, subretinal fluid, RPE atrophy, lipofuscin deposition, drusenoid deposits, pigment epithelial detachment, and choroidal neovascularization in the setting of subretinal hemorrhage with higher sensitivity than clinical examination.[35] OCT may be a more reliable measure of thickness compared to ultrasound for thinner tumors as B scan ultrasound measurements typically include all of the choroid plus retina.[36] Shields et al. found small choroidal melanoma on OCT to exhibit choroidal optical shadowing and overlying choriocapillaris thinning/compaction.[36] The authors also found that, compared to nevi, small choroidal melanomas were more likely to exhibit intraretinal edema ($P=0.003$), “shaggy” photoreceptors or loss of photoreceptors ($P=0.005$), loss of external limiting membrane ($P=0.008$), loss of inner segment-outer segment junction ($P=0.02$), irregularity of the inner plexiform layer ($P=0.04$), and irregularity of the ganglion cell layer ($P=0.04$). Yu et al. noted a correlation between degree of choriocapillaris compression and nevus thickness, which independently elevated risk of choroidal nevus-associated subretinal fluid.[37] The authors found that a lack of choriocapillaris compression was associated with no subretinal fluid. Fluorescein and indocyanine angiography are less helpful than other imaging tests but may display angiographic “hot spots”, leakage suggestive of choroidal neovascularization, and intrinsic tumor vascularization, although tumor vascularization may be easier to identify with larger tumors.

Ultrasonography is the gold standard for monitoring choroidal lesions. A-scan and B-scan ultrasound are indispensable tools for choroidal tumors but are operator dependent and possibly less reliable in thinner tumors. If accurately captured on B-scan, small choroidal

melanomas may exhibit acoustic hallowing and choroidal excavation. On A-scan, tumors may show low to medium reflectivity, similar height of inner tumor spikes or regular decrease in height (positive angle kappa sign), or flickering vertical motion of tumor spikes suggestive of intrinsic vascularity.[14], [38] Except for flat tumors, it is still reasonable to obtain a B-scan/A-scan ultrasound early in serial monitoring.

Artificial Intelligence for Small Choroidal Melanoma: Artificial intelligence (AI) and machine learning have more recently been utilized in the field of ocular oncology and may become a valuable non-invasive method for differentiating a nevus from melanoma, or for differentiating a low-risk and high-risk nevus.[39] AI models would likely rely on 2D fundus photographs which would generally lack the ability to determine tumor thickness, an important risk factor for malignancy, however AI may possibly detect novel biomarkers.[40], [41] Zabor et al. created a classical machine learning model to distinguish small choroidal melanoma from choroidal nevus.[42] The authors used 123 patients diagnosed with a small melanocytic tumor (COMS criteria) as the training set, with 61 patients confirmed as small choroidal melanoma based on either confirmed growth and/or pathology confirmed and 62 lesions as choroidal nevus due to no growth for the past 2 years. Lasso logistic regression was used to select variables differentiating melanoma from nevus in the final model, and this was validated on an external set of 240 patients from a different tertiary center. The “discrimination value” for the model was 0.880 and the top four variables considered important for identifying melanoma were subretinal fluid, height, distance to optic disc, and orange pigment. When tested against the external set, the model distinguished melanoma from nevus with a high discrimination of 0.861. While this important study used variables of interest that were determined a priori, future research will undoubtedly use multimodal imaging without need for inputting variables.

Tumor Biopsy: Because medium and large-sized choroidal melanomas are usually clinically diagnosed, tissue biopsy is typically reserved for tumor prognostication and the patient’s metastatic risk.[43] Indeterminate choroidal lesions, however, would theoretically be the ideal lesions for a diagnostic biopsy. Confirming an ICL is malignant via biopsy and subsequent cytopathology would hasten diagnosis time and lead to earlier treatment.[44] There is reasonable hesitation with thinner tumors as a biopsy may result in insufficient tumor sampling and significant ocular complications particularly in posterior tumors. Therefore, the clinician weighs the risks and benefits of performing a diagnostic biopsy against observing the ICL for documented growth.

Tissue biopsy for posterior uveal tumors has been performed for decades with various techniques described.[45] A trans-scleral approach is typically reserved for tumors anterior to the equator, while a trans-vitreous approach may be utilized for tumors posterior to the equator.[46], [47] Thinner tumors are likely more amenable to a trans-vitreous approach as it affords the surgeon direct visualization of the sample instrument entering the tumor. The two main subtypes for an internal trans-vitreous approach are fine-needle aspiration (FNAB) with a 25-gauge or 27-gauge needle, or vitrectomy-assisted biopsy with a 25-gauge or 27-gauge vitreous cutter.[46] Compared to the 25-gauge cutter, the 27-gauge cutter may have a theoretical advantage for thinner tumors as the cutting port is closer to the tip.[46],

[48] Some studies have demonstrated a higher negative sampling rate in thinner tumors, [49], [50], [51] however Grewal et al. reported a successful sample in a lesion as thin as 0.8 mm.[52] Biopsy complications vary by technique but overall the risks are relatively low.[46] Self-resolving vitreous hemorrhage is the most common complication for a trans-vitreous approach with a much lower risk of retinal detachment and endophthalmitis.[46] A biopsy-associated subretinal hemorrhage has been described but was displaced successfully with intravitreal tPA and C3F8 gas.[53] Assuredly, seeding and extraocular extension are exceedingly rare.[46]

Initial Observation Versus Immediate Treatment: As mentioned previously, one of the most challenging clinical decisions for ICLs are observation and close monitoring for rapid growth, diagnostic biopsy, or immediate treatment. Many experts have weighed the risks and benefits of immediate versus delayed treatment for smaller lesions with “high-risk” features.[54], [55], [56], [57] Even in the SCM subcategory, there is an increasing risk of metastases as the tumor grows. One argument for immediate treatment is the increasing risk of metastases the thicker the lesion, even in the SCM subcategory. The risk of metastases with small choroidal melanoma, the main driver for considering early treatment, appears to be low but is not negligible.[58], [59], [60] Shields et al. analyzed 1780 patients with SCM treated by plaque brachytherapy and found a 10-year metastatic risk of 8.8%.[57] The risk of metastasis increased by every millimeter of thickness (5.9% at 1.1 mm to 2 mm thickness, 8.1% at 2.1 mm to 3 mm thickness, and 13.4% at thicknesses greater than 3 mm). While much lower than the 10-year metastatic risk of medium-sized (26%) and large-sized (49%) choroidal melanomas,[61] the SCM metastatic risk is not trivial and factors into the clinician’s decision-making. The Small Fatal Choroidal Melanoma Study found even a few tumors less than 5 mm in LBD (but larger than 3 mm in LBD) and thickness 1 – 1.7 mm that eventually metastasized – measurements even smaller than included in the COMS small melanoma arm.[62] The same study found SCM patients that metastasized exhibited the same risk factors as lesions of larger sizes including subretinal fluid, thickness greater than 2 mm, visual symptoms, and orange pigment.[62]

Chromosomal abnormalities, somatic mutations, and gene expression profiling (GEP) have provided important diagnostic and prognostic information for UM.[63] Identifying somatic mutations specific to uveal melanoma in a tumor specimen can help aid in diagnosis to confirm uveal origin. The vast majority of UM tumors have activating mutations in GNAQ or GNA11 which ultimately activate downstream signaling pathways.[64][65] Secondary driver mutations may affect BAP1, SF3B1, and EIF1AX which all are generally mutually exclusive of each other with different prognostic values.[66] More somatic mutations can be captured via a next-generation sequencing panel that tests hotspot mutations in the genes GNAQ, GNA11, CYSLTR2, PLCB4, SF3B1, mutations in exons 1–2 of EIF1AX, and all coding exons of the BAP1 gene.

GEP and mRNA expression of the cancer-testis antigen Preferentially Expressed Antigen in Melanoma (*PRAME*) are valuable prognostic tools for metastatic risk. [67], [68], [69][70] GEP is comprised of a 15-gene classification assay. Early studies suggested metastatic risk can be clustered into two highly prognostic GEP subgroups with Class 1 indicating low metastatic risk and Class 2 indicating high metastatic risk.[70] Prior studies evaluating

the relationship between gene expression profiling and tumor size also suggest a small choroidal melanoma by COMS criteria have lower rates of metastases.[71], [72] Larger tumors were more likely to be Class 2 (high risk of metastasis) by GEP compared to smaller tumors.[72], [73] The Collaborative Ocular Oncology Group (COOG) Report 1 was a prospective multicenter trial that demonstrated GEP as the most accurate prognostic marker for metastasis risk compared to clinical features and chromosomal abnormalities.[70] Considering biopsy of indeterminate lesions is relatively safe, sending a sample for GEP may be a viable option when deciding whether to proceed with treatment. Weis et al. analyzed GEP results for 15 high-risk, indeterminate lesions and found 6 patients with Class 1A and 9 patients with Class 1B.[74] None of the Class 1A lesions displayed growth or metastases during a mean 17.1 months of follow-up, and all Class 1B lesions underwent plaque brachytherapy with no report of metastases at the end of the study.[74] Although a small sample size, this study highlights that a biopsy with GEP testing may be an option for patients hesitant with either observation or immediate treatment in ICL. Studies also found *PRAME* to be an independent indicator of increased metastatic risk.[67], [75] Scheffler et al. found *PRAME* expression to be independent of TNM staging, gender, patient age, and tumor thickness.[67] *PRAME* expression was statistically associated with largest basal diameter and GEP status. This study therefore correlated *PRAME* status with certain clinical and molecular features that have been associated with increased metastatic risk. Cai et al. also noted in a retrospective cohort study *PRAME* to be a superior prognostic indicator for metastases compared to TNM staging.

The COOG released an update of the utility of GEP as a prognostic test with Report 2. [76] This prospective multicenter trial assessed 1,577 patients with UM to prospectively validate a prognostic classifier combining the 15-GEP and *PRAME* RNA expression, as well as identify any clinical variables that may enhance the accuracy of the 15-GEP/*PRAME* classifier. The five-year metastasis free survival (MFS) was 95.6% for class 1/*PRAME*(-), 80.6% for class 1/*PRAME*(+), 58.3% for class 2/*PRAME*(-), and 44.8% for class 2/*PRAME*(+).[76] Tumor diameter, not thickness, was the only clinical variable that demonstrated additional prognostic value. There was no significant difference in MFS or melanoma-specific survival (MSS) between Class 1A and Class 1B – therefore the two groups were combined for analysis. The findings of the study support a new 4-group 15-GEP/*PRAME* system, and suggest the combination to have a higher prognostic value than other chromosomal prognostic markers.[76]

Treatment

When a tumor is confirmed to be malignant (either by rapid growth or diagnostic biopsy) and characterized as a small choroidal melanoma, then treatment is recommended and should not be delayed.[54], [55], [56], [57], [58], [59] Strict guidelines regarding appropriate length of observation do not exist, however a reasonable approach may be to not delay treatment if the clinician is convinced the lesion is malignant.

Treatment options for SCM are largely globe-salvaging therapies and include laser therapy or radiation.[77] Transpupillary thermotherapy (TTT) was first described in 1995 as an alternative to radiation therapy for choroidal melanoma and is performed with a 810 nm

diode laser to the tumor and 1 mm of surrounding tissue.[78], [79] The mechanism of action is due to hyperthermia of the tumor cells leading to a cytotoxic effect with eventual tumor necrosis, and the technique may be employed on tumors up to 4 mm in thickness.[79] Shields et al. reported 256 cases with newly diagnosed choroidal melanoma treated with TTT and noted a tumor recurrence rate of 9% with a relatively short follow-up duration.[80] A more recent retrospective study demonstrated a high Kaplan-Meier estimate for tumor recurrence and correlated this with the number of high-risk tumor features.[81] Extraocular extension was associated with eyes that underwent additional TTT sessions after intraocular tumor recurrence, possibly due to residual intrascleral melanoma.[81] Due to the risk of recurrence, TTT is generally reserved for either select cases of lower-risk melanocytic tumors or as an adjunct to radiation therapy. Photodynamic therapy (PDT), widely used for retinal and choroidal vascular lesions in ocular oncology, has been studied as a possible treatment for SCM given its limited invasiveness compared to radiation therapy.[82] Multiple antitumoral effects of PDT may be present including cytotoxic effects on tumor cells, activation of a local immune response, and destruction of peritumoral vasculature.[83] Some reports suggest an increased efficacy in amelanotic lesions compared to melanotic ones, possibly due to cytoprotection by increased melanosomes in the RPE or uveal tract. [83], [84] Roelofs et al. noted a 5-year treatment-success rate at 38.4% for predominantly pigmented tumors, while a meta-analysis by Yordi et al. of 7 studies with PDT as primary treatment reported a wide response rate of 58–100%.[82], [83] Although there were variable treatment regimens and follow-up durations, the authors of the meta-analysis suggest that PDT may be considered in select cases of small, amelanotic lesions.[82]

Radiation therapy is the most common treatment for choroidal melanoma and can be offered as plaque brachytherapy or proton beam radiotherapy. While COMS did not treat small choroidal melanoma with brachytherapy at enrollment, 67 patients required treatment at follow-up and melanoma-specific mortality at 5 years was 1%.[58] More recent retrospective studies reported Iodine-125 brachytherapy provided excellent local tumor control with low rates of metastases.[57], [60] Yipari et al. reported a retrospective study of 161 cases of small choroidal melanoma (tumor apical height of 1.0 – 2.5 mm and LBD < 16 mm) treated with I-125 brachytherapy plaque (apical height dose of 85 Gy).[60] Only one recurrence event (1%) and one enucleation event (1%) was observed. 69% of patients maintained 20/50 vision or better and no metastatic events were observed at 3 years. Proton beam radiotherapy has similarly demonstrated acceptable local tumor control in patients with small choroidal melanoma with low rates of metastases.[85] Similar to plaque brachytherapy, proton beam irradiation also demonstrated decreased visual acuity over time, with over half the eyes at 20/80 or worse at 5 years.[85] The beta-emitter Ruthenium-106 is more commonly used in Europe and has a more limited depth of penetration compared to Iodine-125 which may portend a benefit in thinner tumors.[86], [87] When including tumors with maximum apical height of 6.5 mm, Cennamo et al. treated 355 eyes with Ruthenium-106 and reported 38% of eyes with radiation maculopathy, and cancer-free survival of 99%, 97%, and 85% at 5, 7, and 9 years respectively.[87] Palladium-103 has also been utilized for the treatment of choroidal melanoma, with authors reporting the lower energy photons from this isotope are more rapidly absorbed by the tumor and less likely to reach most normal structures compared to Iodine-125.[88], [89] Semenova and

Finger published a retrospective case series of 72 patients with small choroidal melanoma treated with Palladium-103.[90] After a mean 54 months of observation, 100% of patients achieved local tumor control and there were no melanoma-related metastases.[90] 43.1% of the cases developed radiation maculopathy and 94.4% maintained vision of 20/200 or better. A newer radiation treatment being evaluated for small choroidal melanoma is Yttrium-90 disc brachytherapy in which a 6 mm radioactive disc in a titanium shell is placed on the sclera for several minutes depending on the dosage required.[91] Long-term tumor control rates are still being investigated.

Despite excellent local tumor control, radiotherapy may result in a poor visual prognosis especially if the tumor is posterior. A novel treatment option in clinical trial for small choroidal melanoma is the photoactivated therapy belzupacap sarotalocan (bel-sar; AU-011), a first-in-cancer virus-like nanoparticle with the goal of treating the tumor while preserving visual acuity.[92] Viral-like particle bioconjugates (VPBs) are injected either intravitreally or via the suprachoroidal space and bind preferentially to heparan sulfate proteoglycans that are overexpressed by uveal melanoma cells. The VPBs are conjugated with a photosensitizer which is activated by a 689-nm near-infrared laser leading to both cytotoxic effects of the tumor and T-cell immune activation. Tumor criteria are thickness between 0.5 mm and 2.5 mm, largest basal diameter less than 10 mm, and documented growth within two years of screening. Interim Phase 2 results demonstrated excellent tumor control rate (89–100%) with a high visual acuity preservation rate.[93] A Phase 3 randomized controlled trial is currently enrolling and will employ a suprachoroidal route for drug delivery.

Conclusion

Small choroidal melanoma remains an important subtype of uveal melanoma that warrants further optimization in diagnosis and treatment. Despite improvements in diagnostic imaging and molecular testing, indeterminate choroidal lesion management is controversial. A small choroidal lesion with several high-risk features but no growth over many years may be appropriately designated as benign, but could just as well undergo treatment if the duration were unknown. While heavy drusen overlying a lesion suggests chronicity, sudden growth would indicate malignant transformation and warrant treatment. These scenarios, among countless others, add to the controversy of appropriate management for SCM. Balancing the risk of visual morbidity and metastases (albeit low risk) regarding treatment timing remains a challenging discussion, however new advances in diagnostic imaging, molecular testing, artificial intelligence, and vision-preserving therapy may hopefully decrease current controversy.

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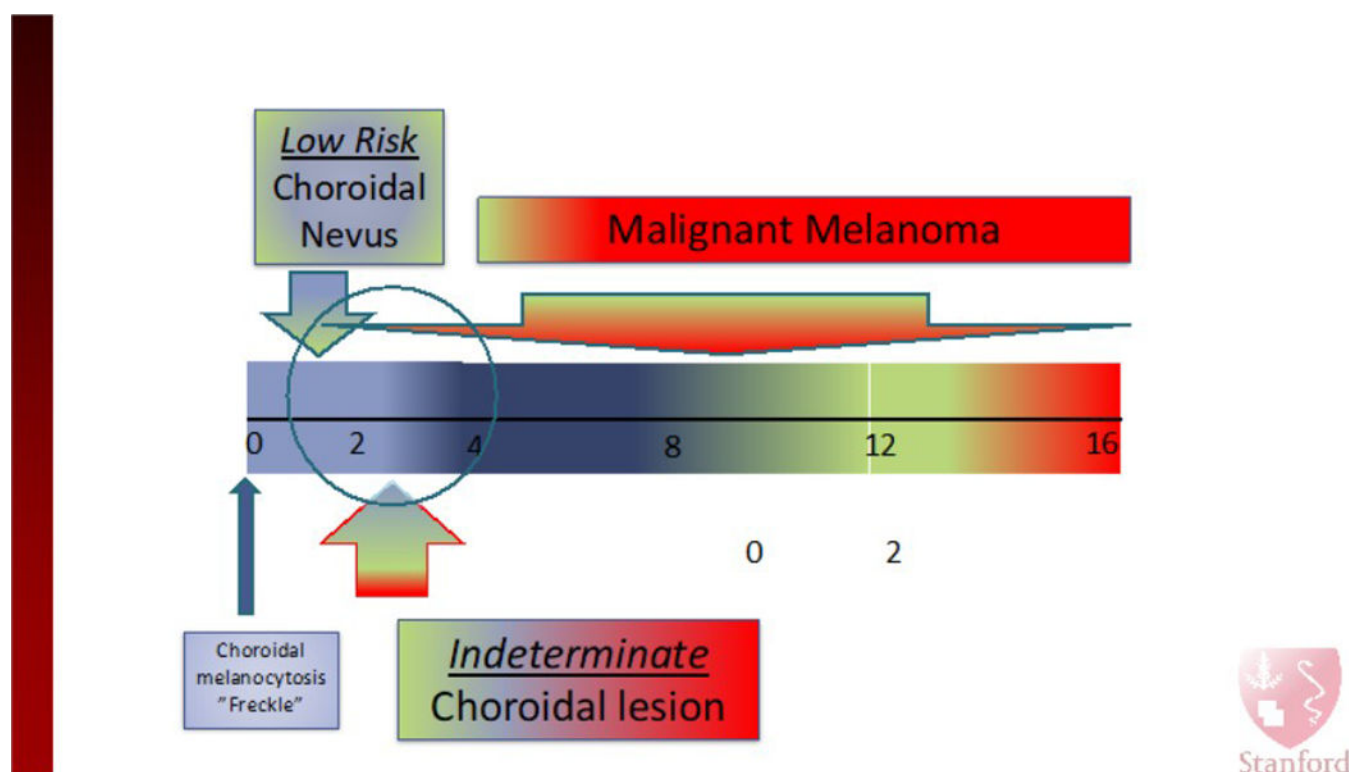


Figure 1:
A schematic representation of melanotic choroidal lesions and their presumed diagnosis based on tumor thickness (x-axis).

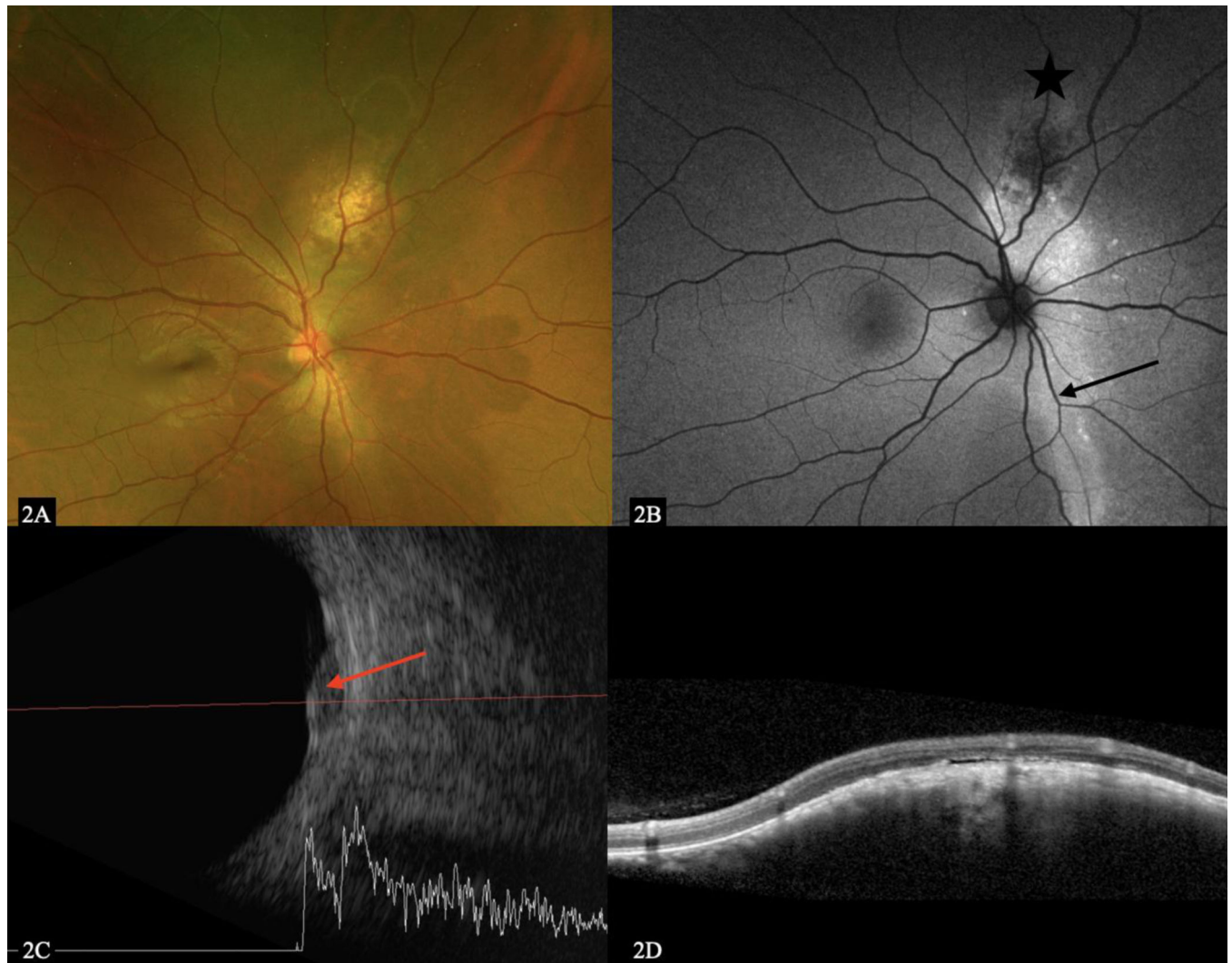


Figure 2:

Multimodal imaging of an indeterminate, amelanotic choroidal lesion of the right eye. 2A: A cropped ultra-widefield pseudocolor photo demonstrates an amelanotic, elevated lesion superonasal to the optic nerve. 2B: Fundus autofluorescence shows a hypo-autofluorescent lesion (black star) with guttering hyper-autofluorescence corresponding to subretinal fluid (black arrow). 2C: B-scan ultrasound of the lesion (red arrow) shows a medium-low reflective choroidal lesion. 2D: OCT of the lesion demonstrates a deep choroidal lesion with overlying choriocapillaris compaction, outer retinal thinning, and a sliver of subretinal fluid. This lesion was observed with no signs of growth on follow-up.

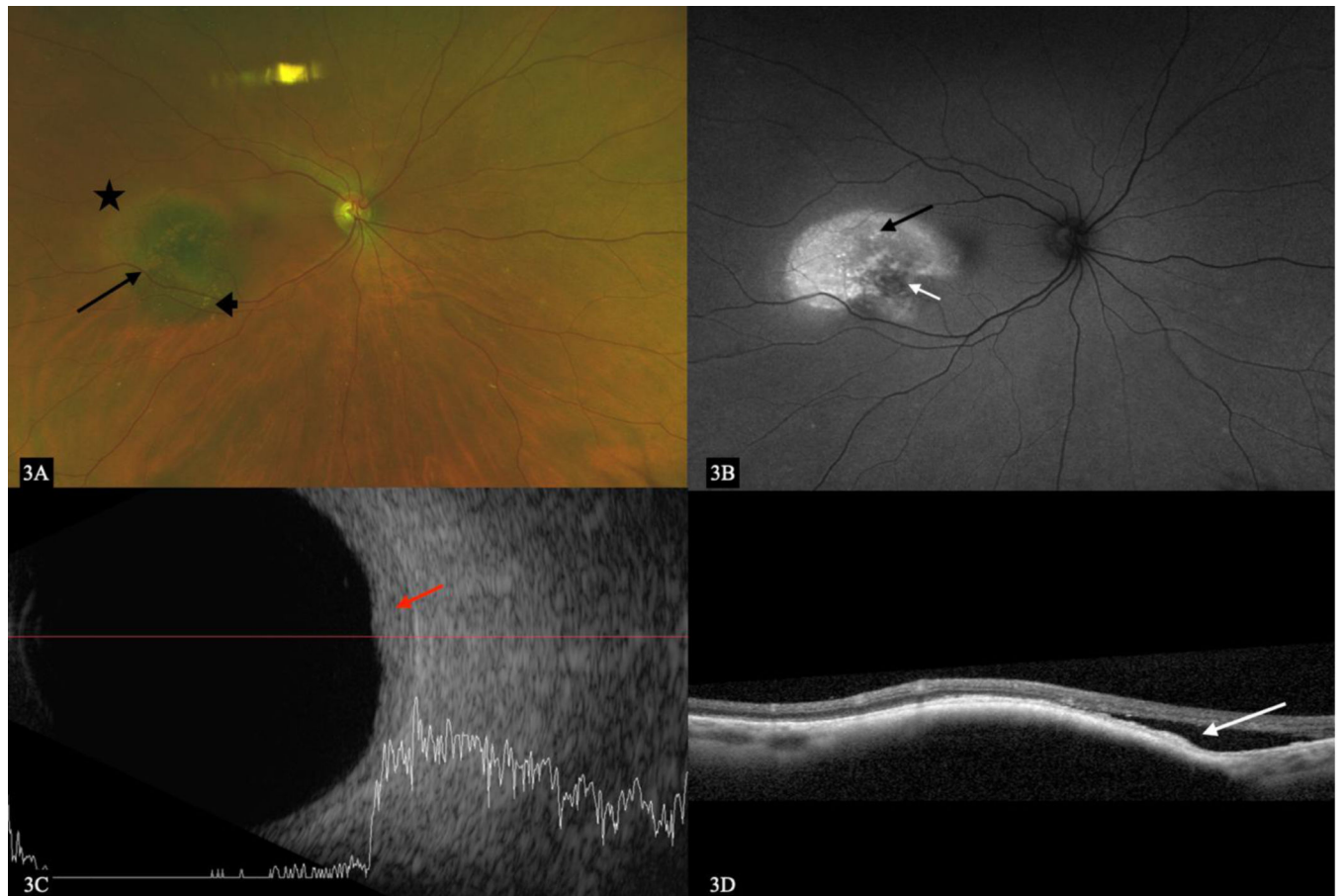


Figure 3:

Multimodal imaging of an indeterminate choroidal lesion of the right eye. 3A: A cropped ultra-widefield pseudo-color photo demonstrates a temporal melanotic elevated lesion with subretinal fluid (black star), orange pigment (long black arrow), and drusen (short black arrow). 3C: Fundus autofluorescence shows a largely hyper-autofluorescent lesion with focal hyper-autofluorescent spots (black arrow) suggesting lipofuscin accumulation and areas of hypo-autofluorescence (white arrow) suggesting retinal pigment epithelium atrophy. 2C: B-scan ultrasound of the elevated lesion (red arrow) shows iso-reflectivity relative to the rest of the choroid. 2D: OCT of the lesion demonstrates a deep choroidal lesion with overlying choriocapillaris compaction, outer retinal thinning, and subretinal fluid (white arrow). This lesion was observed with no signs of growth on follow-up.